

Andrew Zagula

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EDUCATION

California Institute of Technology

B.S. in Computer Science, Minors in Robotics

Pasadena, CA

Expected Jun 2029

EXPERIENCE

FinalDose (YC P26)

Software Engineer

Remote

Aug 2026 – Present

- Built LLM-agnostic 32-tool MCP server for DNA parsing, sequence alignment, lab-procedure simulation, and live record/file access, backed by 28-function Python/Biopython compute library shared with 8 REST endpoints
- Engineered lab data platform (Django, PostgreSQL, TypeScript/Next.js) on AWS ECS via Docker/Terraform, domain-restricted Google Workspace SSO, per-experiment write access, and unified Google Drive/S3 file storage

Terranox AI (YC W26)

Software Engineer

San Francisco, CA

Jun 2026 – Aug 2026

- Shipped land evaluation dashboard (TypeScript/Next.js) over 8,000 parcels, spatially querying a 200K-file AWS RDS/PostGIS index over S3-backed survey data across table, 2D map, 3D layer-stack, and heat map grid views
- Designed LLM scoring pipeline (Python, EC2) ranking 8,000 parcels by owner sell likelihood, parsing 150 public companies' financial filings into normalized weighted scores over cash runway, owner type, and resale history

University of California, Berkeley

Student Researcher

Remote

Jul 2024 – Dec 2025

- First-authored AutoAdv (NeurIPS Lock-LLM 2025), an automated adversarial prompting framework for multi-turn LLM jailbreaking; achieved 99% attack success rate against Qwen3-235B in 6 turns, +24% vs. single-turn (75%)
- Implemented two-phase prompt rewriting pipeline with attacker LLM Grok 3 Mini and GPT-4o mini judge; developed 4 sampling temperature strategies and cataloged 28 effective rewriting techniques to guide future attacks

Boston University

Research Intern (RISE)

Boston, MA

Jun 2025 – Aug 2025

- Co-authored BabyVLM-V2 (CVPR 2026), a developmentally inspired framework for sample-efficient vision-language models; constructed DevCV Toolbox, a 10-task benchmark generated from 478 hours of infant-perspective video data
- Developed OpenCV pipeline with frame sampling and sliding-window segmentation; applied YOLOE-11 for object detection and tracking, filtering 2.2K+ usable clips (~63% yield) from 3.5K+ candidates via Label Studio

PROJECTS

PaperTrail | *Python, FastAPI, Next.js, TypeScript, LangChain, LangGraph, ChromaDB, SQLite*

- Built research assistant with arXiv URL/PDF ingestion, LLM-generated multi-query paper discovery, structured paper breakdowns, and 3 LangGraph workflows for comparison, ideation, and implementation planning
- Developed provider-agnostic RAG pipeline using LangChain for multi-turn Q&A with section-level citations; integrated ChromaDB top-5 retrieval and SQLite storage for papers, chats, and generated research artifacts

AUDIT | *Python, OpenAI API, ChromaDB, SQLite, Typer, PyInstaller, npm*

- Built npm-installable CLI security scanner detecting repository vulnerabilities via retrieval over a 50-pattern CWE/OWASP knowledge base, generating JSON reports with confidence scores and remediation guidance
- Engineered scanning pipeline with AST and regex function chunking, weighted heuristic prefiltering, ChromaDB semantic retrieval, parallel LLM review, and content-hash SQLite caching, reducing LLM reviews by ~70%

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, TypeScript, Java, SQL

Frameworks: React, Next.js, Node.js, FastAPI

Developer Tools: Git, Docker, PostgreSQL, Firebase, SQLite, ChromaDB

Libraries: PyTorch, Hugging Face, NumPy, Pandas, OpenCV, Tailwind CSS

AWARDS

- 2x USAMO Qualifier, 5x AIME Qualifier, USA Computing Olympiad Gold, NY Tech Week \$1K Hackathon Winner